JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI ARUL MARIAMMAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

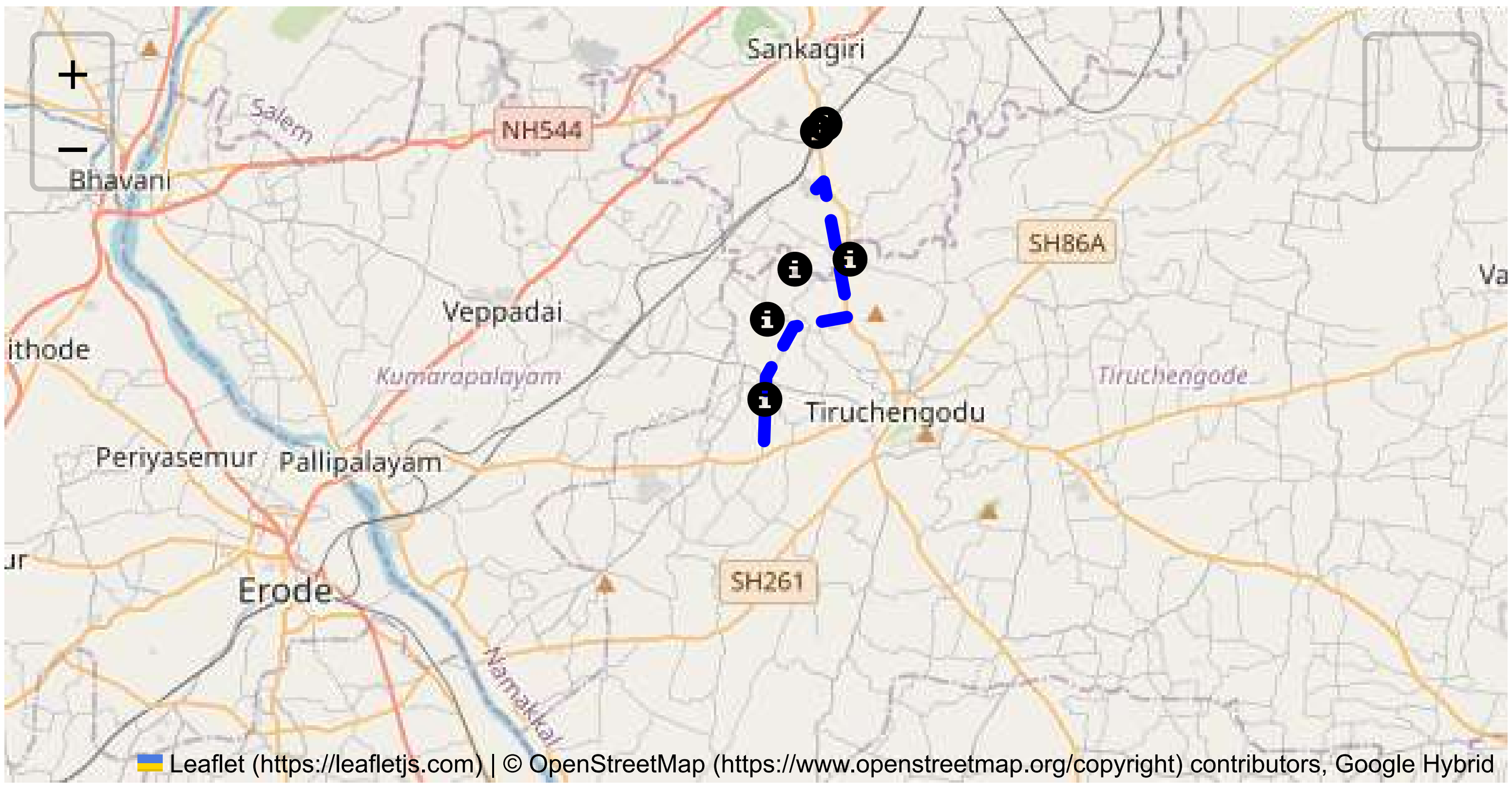
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

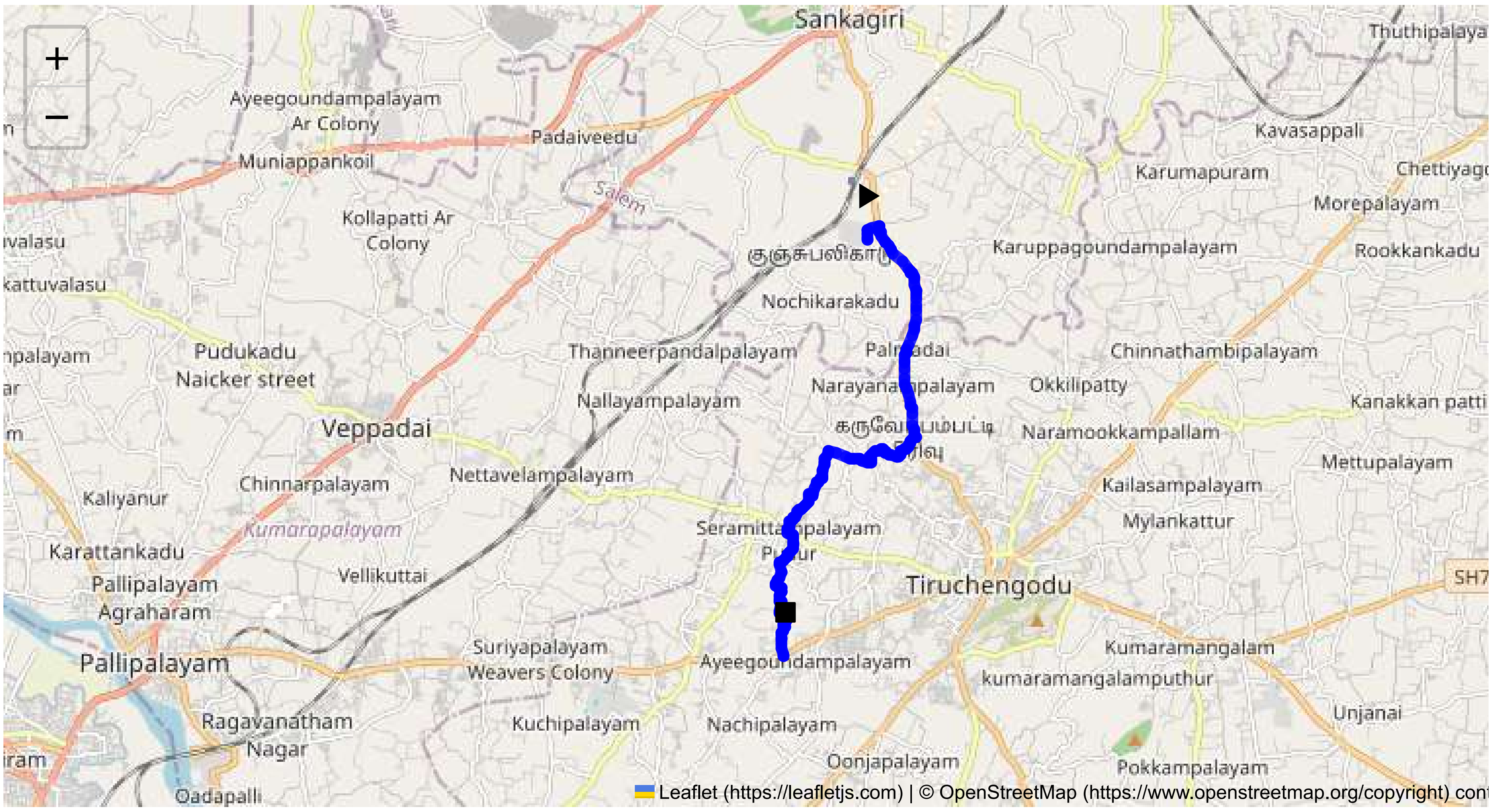
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 11.54 km
Estimated Duration: 0.4 hours
Adjusted Duration (Heavy Vehicle): 0.5 hours
Start: (11.4381, 77.8734)
End: (11.366165, 77.858658)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, to 9V85+V39, Varagurampatti, spans approximately 11.54 kilometers, passing through the key points: Kangeyampalayam, Seetharampalayam, Seenevasampalayam, and Devanankurichi. It is primarily a rural route with segments of local roads connecting these small towns and villages.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences tropical weather conditions. The monsoon season, from June to September, can bring heavy rains leading to road flooding and reduced visibility. The summer months can be quite hot, potentially affecting vehicle performance.

3. Analysis of Traffic Patterns

Traffic tends to peak during local rush hours, approximately 8:00-10:00 AM and 5:00-7:00 PM. The areas around Tiruchengode can experience congestion, particularly as local businesses open and close. Off-peak hours usually see less traffic, but agricultural vehicles are common during harvest season.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route:

- Some segments may have potholes and uneven surfaces, particularly after the monsoon season.
- The infrastructure, including signage and road markers, may be sparse in rural areas.
- Narrow roads can limit maneuverability, especially for heavy vehicles.

5. Suggestions for Alternative Routes

In case of emergencies:

- Consider bypass routes connecting through nearby NH-44 or SH-86 for quicker access to major highways.
- Detour options should be pre-planned using local bypasses to avoid congestion.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is regulated under Indian road safety and national transportation laws. Particular routes might have restrictions not clearly marked, so consulting local authorities for specific approvals is advised.

7. Overview of Historical Incidents

While detailed records are not available, rural routes in Tamil Nadu are generally less prone to incidents but can suffer impacts from events like vehicle overturns or spillage due to road quality issues and weather conditions.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural zones. It's important to minimize risks of hazardous spills due to potential contamination. Livelihoods often depend on these lands, so environmental precautions are vital.

9. Analysis of Communication Coverage

Rural and semi-rural areas often have spotty communication coverage. Certain sections near Seenevasampalayam and Devanankurichi might experience signal drops. Using a reliable GPS system with offline capabilities is recommended.

10. Estimated Emergency Response Times

Emergency response can vary greatly:

- Densely populated areas near Tiruchengode might see response times of 20-30 minutes.
- Rural segments likely to experience longer response times, potentially over an hour due to distance and fewer facilities.

12. Overall Summary of Risk Assessment

The route presents moderate risks primarily due to weather variability, road conditions, and communication limitations. Planning for adverse weather, ensuring vehicle maintenance, and coordinating with local authorities for hazardous material guidelines are essential steps in mitigating potential challenges. Regular updates on road conditions and routing can further enhance safety. Emergency preparedness is crucial, especially in rural segments where response times are longer.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.40413, 77.88197	10 KM/Hr
5	Turn	Medium	11.40070, 77.87400	30 KM/Hr
6	Blind Spot	Blind Spot	11.39972, 77.87402	10 KM/Hr
7	Turn	Medium	11.39975, 77.87396	30 KM/Hr
8	Turn	Medium	11.39980, 77.87261	30 KM/Hr
9	Turn	Medium	11.40017, 77.87226	30 KM/Hr
10	Turn	Medium	11.40031, 77.87182	30 KM/Hr
11	Turn	High	11.40161, 77.86664	15 KM/Hr
12	Turn	Medium	11.39973, 77.86588	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	High	11.39963, 77.86590	15 KM/Hr
14	Turn	High	11.39952, 77.86610	15 KM/Hr
15	Turn	Medium	11.39707, 77.86556	30 KM/Hr
16	Turn	High	11.39624, 77.86468	15 KM/Hr
17	Turn	High	11.39405, 77.86394	15 KM/Hr
18	Turn	Medium	11.39402, 77.86373	30 KM/Hr
19	Turn	High	11.39411, 77.86328	15 KM/Hr
20	Turn	Medium	11.38934, 77.85988	30 KM/Hr
21	Turn	High	11.38765, 77.85935	15 KM/Hr
22	Turn	High	11.38701, 77.86036	15 KM/Hr
23	Turn	Medium	11.38269, 77.85866	30 KM/Hr
24	Turn	High	11.38262, 77.85864	15 KM/Hr
25	Turn	Medium	11.38244, 77.85815	30 KM/Hr
26	Turn	Medium	11.37945, 77.85787	30 KM/Hr
27	Turn	Medium	11.37928, 77.85763	30 KM/Hr
28	Turn	High	11.37859, 77.85734	15 KM/Hr
29	Turn	High	11.37809, 77.85828	15 KM/Hr
30	Turn	Medium	11.37794, 77.85829	30 KM/Hr
31	Turn	Medium	11.37767, 77.85815	30 KM/Hr
32	Turn	Medium	11.37731, 77.85832	30 KM/Hr
33	Turn	High	11.37578, 77.85794	15 KM/Hr
34	Turn	Medium	11.37573, 77.85800	30 KM/Hr
35	Turn	High	11.37567, 77.85848	15 KM/Hr
36	Turn	High	11.37559, 77.85853	15 KM/Hr
37	Turn	Medium	11.37396, 77.85794	30 KM/Hr
38	Turn	Medium	11.37385, 77.85798	30 KM/Hr
39	Turn	High	11.37292, 77.85935	15 KM/Hr
40	Turn	Medium	11.37183, 77.85882	30 KM/Hr
41	Turn	Medium	11.37076, 77.85882	30 KM/Hr
42	Turn	High	11.36644, 77.85867	15 KM/Hr

Route Photos of Risky Spots

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.40413, 77.88197

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.40070, 77.87400

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.39972, 77.87402

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39975, 77.87396

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39980, 77.87261

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.40017, 77.87226

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.40031, 77.87182

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40161, 77.86664

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39973, 77.86588

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39963, 77.86590

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39952, 77.86610

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39707, 77.86556

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39624, 77.86468

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39405, 77.86394

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39402, 77.86373

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39411, 77.86328

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38934, 77.85988

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38765, 77.85935

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38701, 77.86036

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38269, 77.85866

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38262, 77.85864

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38244, 77.85815

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37945, 77.85787

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37928, 77.85763

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37859, 77.85734

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37809, 77.85828

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37794, 77.85829

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37767, 77.85815

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37731, 77.85832

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37578, 77.85794

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37573, 77.85800

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37567, 77.85848

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37559, 77.85853

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37396, 77.85794

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37385, 77.85798

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37292, 77.85935

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37183, 77.85882

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37076, 77.85882

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36644, 77.85867
